Diffie-Hellman Key Exchange
and
Man-in-the-middle Attack

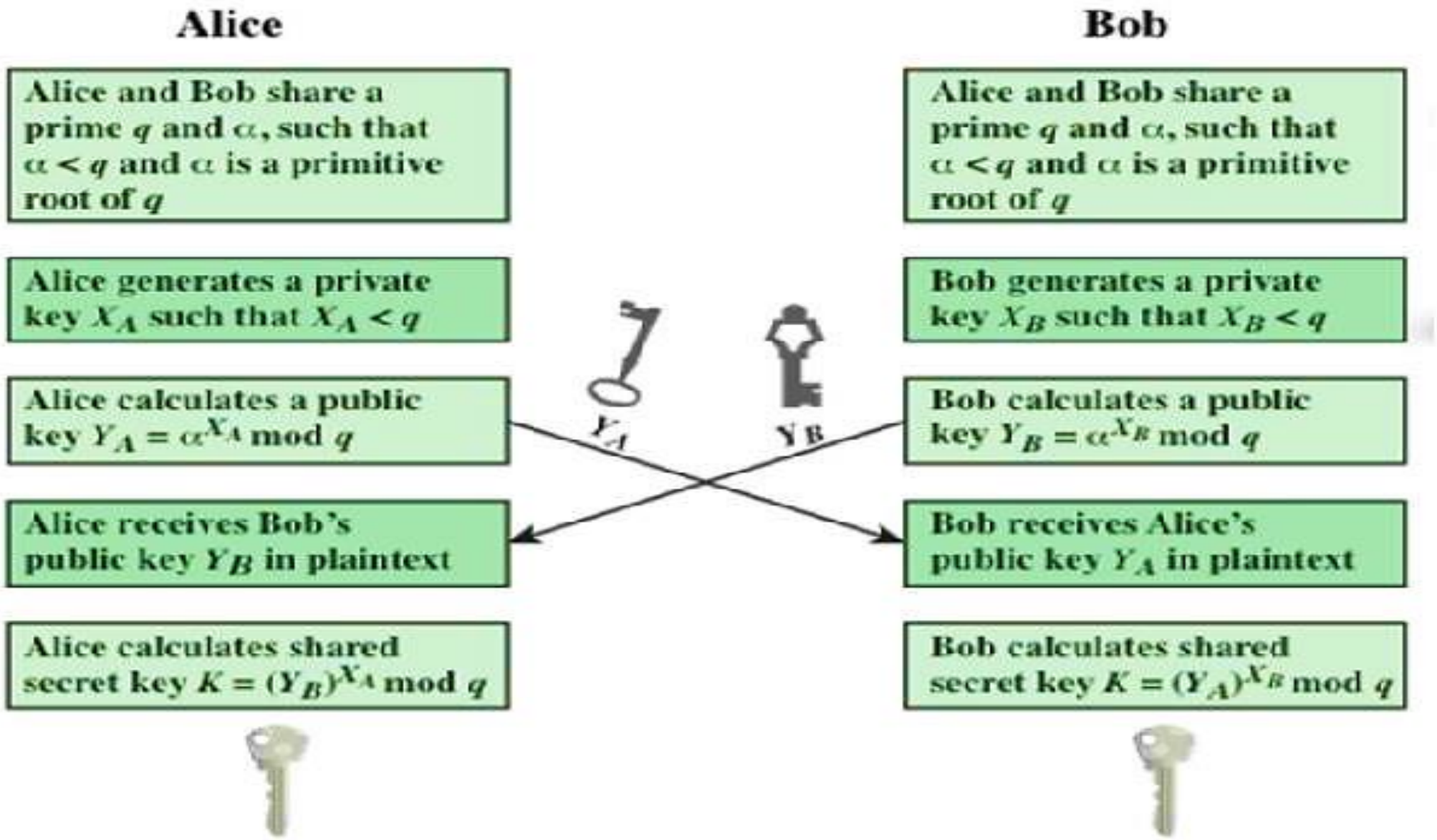
Introduction to D-H Key Exchange

- first public-key type scheme proposed
- by Diffie & Hellman in 1976 along with the exposition of public key concepts
  - note: now know that Williamson (UK CESG) secretly proposed the concept in 1970
- is a practical method for public exchange of a secret key
- used in a number of commercial products

Introduction to D-H Key Exchange … Contd.

- a public-key distribution scheme
  - cannot be used to exchange an arbitrary message
  - rather it can establish a common key
  - known only to the two participants
- value of key depends on the participants (and their private and public key information)
- based on exponentiation in a finite (Galois) field (modulo a prime or a polynomial) - easy
- security relies on the difficulty of computing discrete logarithms (similar to factoring) – hard

Introduction to D-H Key Exchange … Contd.



Diffie-Hellman Setup

- all users agree on global parameters:
  - large prime integer or polynomial q
  - α being a primitive root mod q
- each user (eg. A) generates their key
  - chooses a secret key (number): $x_A < q$
  - compute their public key: $y_A = \alpha^{x_A} \bmod q$
- each user makes public that key y_A

Diffie-Hellman Setup … Contd.

- shared session key for users A & B is K_{AB}:

  $K_{AB} = \alpha^{x_A . x_B} \bmod q$

  $= y_A^{x_B} \bmod q$ (which **B** can compute)

  $= y_B^{x_A} \bmod q$ (which **A** can compute)
- K_{AB} is used as session key in private-key encryption scheme between Alice and Bob
- if Alice and Bob subsequently communicate, they will have the **same** key as before, unless they choose new public-keys
- attacker needs an x, must solve discrete log

Diffie-Hellman Example

- users Alice & Bob who wish to swap keys:
- agree on prime q=353 and α=3
- select random secret keys:
  - A chooses x_A=97, B chooses x_B=233
- compute respective public keys:
  - $y_A=3^{97} \bmod 353 = 40$ (Alice)
  - $y_B=3^{233} \bmod 353 = 248$ (Bob)
- compute shared session key as:
  - $K_{AB}= y_B^{x_A} \bmod 353 = 248^{97} = 160$ (Alice)
  - $K_{AB}= y_A^{x_B} \bmod 353 = 40^{233} = 160$ (Bob)

Diffie-Hellman – Multiparty Computation

Steps

1. The parties agree on the algorithm parameters p and g.
2. The parties generate their private keys, named a, b, and c.
3. Alice computes $g^a \bmod p$ and sends it to Bob.
4. Bob computes $(g^a)^b \bmod p = g^{ab} \bmod p$ and sends it to Carol.
5. Carol computes $(g^{ab})^c \bmod p = g^{abc} \bmod p$ and uses it as her secret.
6. Bob computes $g^b \bmod p$ and sends it to Carol.
7. Carol computes $(g^b)^c \bmod p = g^{bc} \bmod p$ and sends it to Alice.
8. Alice computes $(g^{bc})^a \bmod p = g^{bca} \bmod p = g^{abc} \bmod p$ and uses it as her secret.
9. Carol computes $g^c \bmod p$ and sends it to Alice.
10. Alice computes $(g^c)^a \bmod p = g^{ca} \bmod p$ and sends it to Bob.
11. Bob computes $(g^{ca})^b \bmod p = g^{cab} \bmod p = g^{abc} \bmod p$ and uses it as his secret.

Key Exchange Protocols

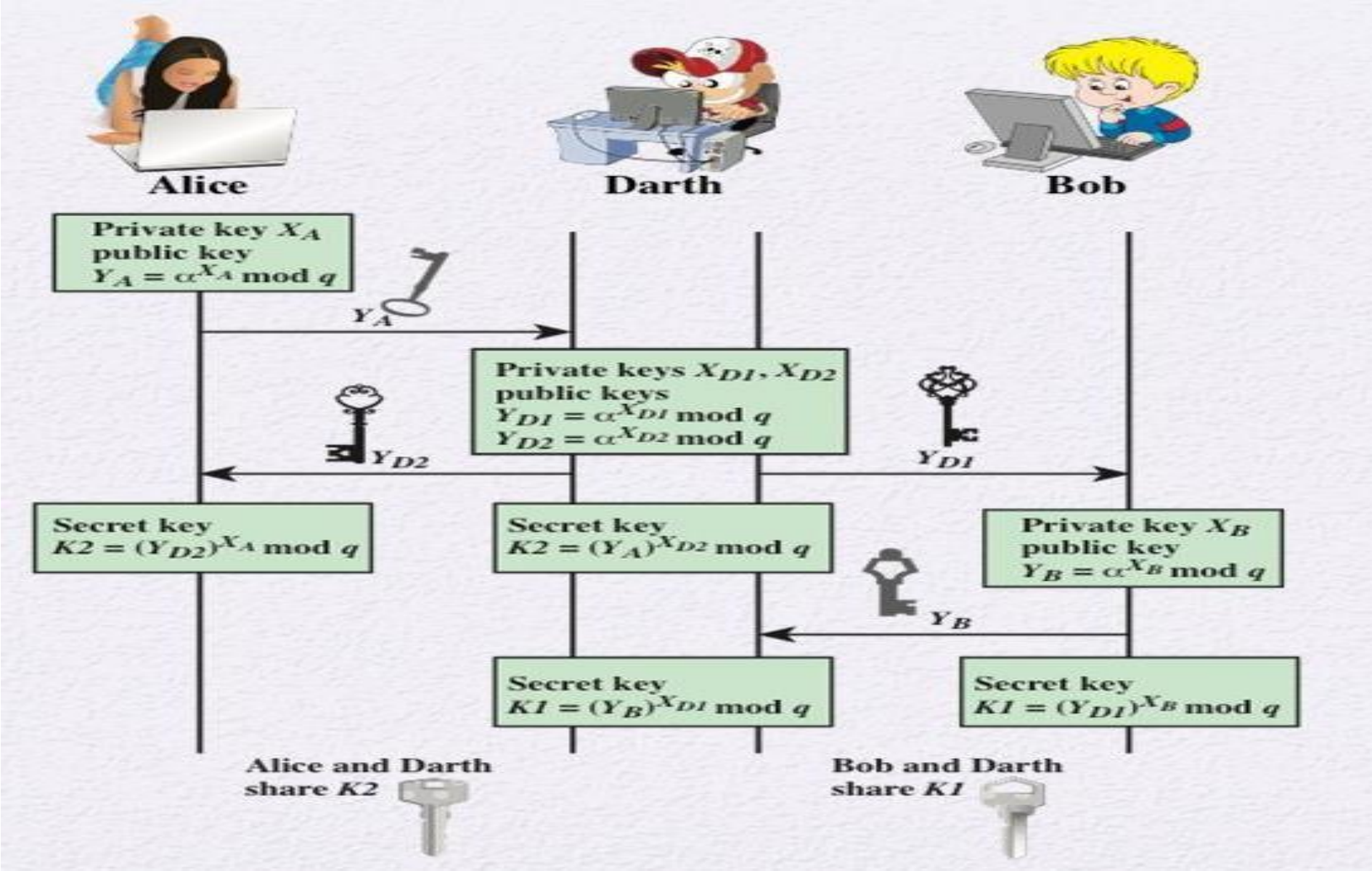
- users could create random private/public D-H keys each time they communicate
- users could create a known private/public D-H key and publish in a directory, then consulted and used to securely communicate with them
- both of these are vulnerable to a meet-in-the-middle attack
- authentication of the keys is needed

Introduction to Man-in-the-Middle Attack

1. Darth prepares by creating two private / public keys
2. Alice transmits her public key to Bob
3. Darth intercepts this and transmits his first public key to Bob. Darth also calculates a shared key with Alice
4. Bob receives the public key and calculates the shared key (with Darth instead of Alice)
5. Bob transmits his public key to Alice
6. Darth intercepts this and transmits his second public key to Alice. Darth calculates a shared key with Bob
7. Alice receives the key and calculates the shared key (with Darth instead of Bob)

Darth can then intercept, decrypt, re-encrypt, forward all messages between Alice & Bob

Introduction to Man-in-the-Middle Attack … Contd.



Practice Problems

1) Alice and Bob use the Diffie-Hellman key exchange technique with a common prime q=353 and a primitive root α=3.
   a) If Alice has a private key X_A=97, find her public key Y_A.
   b) If Bob has a private key X_B=233, find his public key Y_B.
   c) What is the shared secret key between Alice and Bob?
2) Alice and Bob use the Diffie-Hellman key exchange technique with a common prime q=23 and a primitive root α=5.
   a) If Bob has a public key Y_B=10, what is Bob's private Y_B?
   b) If Alice has a public key Y_A=8, what is the shared key K with Bob?
   c) Show that 5 is a primitive root of 23.